

Random Signal Theory–EEE 554

Lecture Notes 3

Transformations of Two Jointly Distributed Random Variables

[Reading: Kay, Sections 7.6, 12.6]

Example:

Find $\Pr(X + Y \leq 1)$ given the joint PDF of X and Y as (pages 7-11)

$$f_{XY}(x, y) = \begin{cases} 2e^{-x}e^{-y}, & 0 \leq y \leq x < \infty \\ 0, & \text{otherwise} \end{cases}$$

and their sample space as $S_{X,Y} = \{(x, y) : 0 \leq x < \infty, y \leq x\}$

By letting $Z = X + Y$,

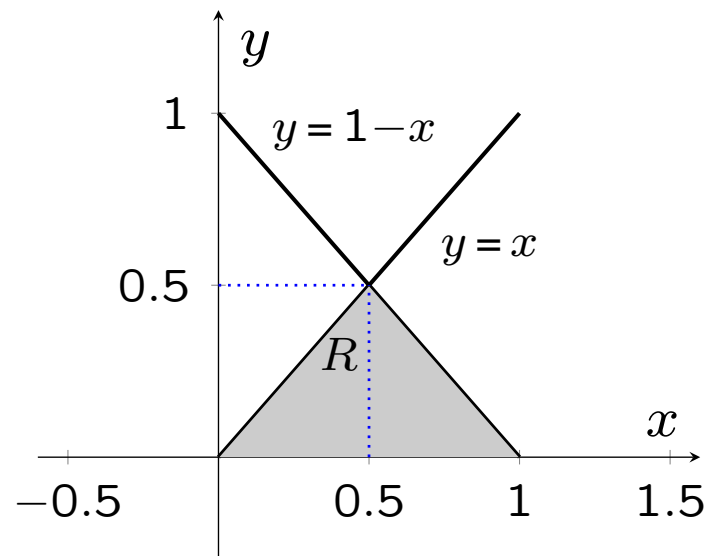
$$\Pr(X + Y \leq 1) = \Pr(Z \leq 1) = \Pr((x, y) \in R) = \int \int_R f_{XY}(x, y) \, dx \, dy$$

$\Rightarrow$ Must find (x, y) points in region R

Region R includes (x, y) points from $S_{X,Y}$ such that $x + y \leq 1$ or $y \leq 1 - x$

$\Rightarrow R$ includes (x, y) points in the **intersection** of $S_{X,Y}$ and $y \leq 1 - x$

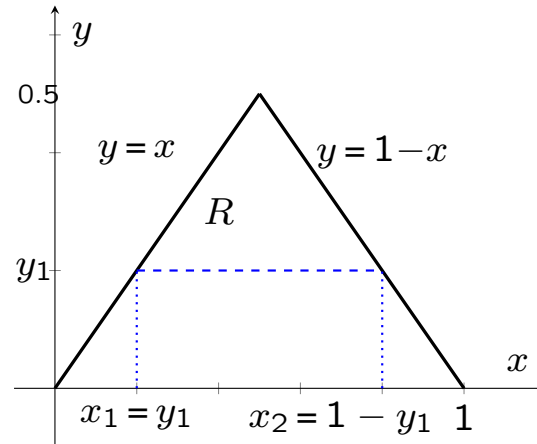
$\Rightarrow R = \left\{ (x, y) : 0 \leq x \leq \infty, y \leq x \right\} \cap \left\{ (x, y) \in S_{X,Y} : x + y \leq 1 \right\}$ (shaded region)



Note: to find point of intersection of the two lines,

$$\text{let } y = x \text{ in } y = 1 - x \Rightarrow x = 1 - x \Rightarrow x = 0.5 = y$$

Need to find limits of integration of $\iint_R f_{XY}(x, y) \, dx \, dy = \Pr(X + Y \leq 1)$



Outer integral limits (in terms of y): $y = 0$ to 0.5

Inner integral limits (in terms of x):

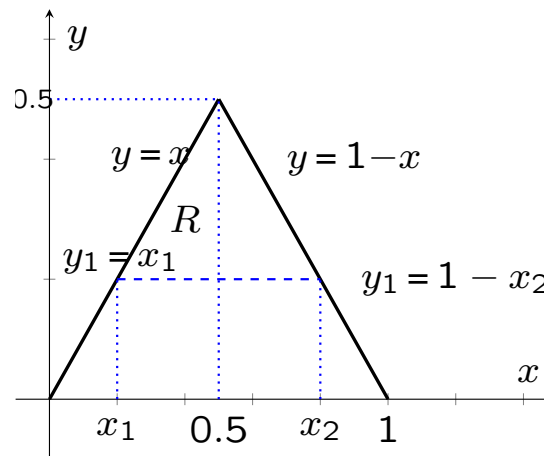
fix $y = y_1$, then range of x in shaded region are $x_1 = y_1$ to $x_2 = 1 - y_1$

$$\Pr(X + Y \leq 1) = \int_{y=0}^{0.5} \left[\int_{x=y}^{1-y} f_{XY}(x, y) \, dx \right] dy$$

Substitute for the known joint PDF of X and Y

$$\begin{aligned}\Pr(X + Y \leq 1) &= \int_{y=0}^{0.5} \int_{x=y}^{1-y} f_{XY}(x, y) \, dx \, dy \\&= \int_{y=0}^{0.5} \int_{x=y}^{1-y} 2e^{-x} e^{-y} \, dx \, dy \\&= 2 \int_{y=0}^{0.5} e^{-y} \left[\int_{x=y}^{1-y} e^{-x} \, dx \right] dy \\&= 2 \int_{y=0}^{0.5} e^{-y} \left[e^{-y} - e^{-(1-y)} \right] dy \\&= 2 \int_{y=0}^{0.5} (e^{-2y} - e^{-1}) \, dy \\&= (-e^{-1} - e^{-1}) - (-1) = 1 - 2e^{-1}\end{aligned}$$

Alternative solution: same answer can be obtained if inner integral is in terms of y



Integration in two steps:

For $0 \leq x < 0.5$ limits, fix $x = x_1$, then y limits are from 0 to $y_1 = x_1$

For $0.5 \leq x < 1$ limits, fix $x = x_2$ then y limits are from 0 to $y_1 = 1 - x_1$

$$\begin{aligned} \Pr(X + Y \leq 1) &= \int_{x=0}^1 \int_R f_{XY}(x, y) \, dy \, dx = \int_{x=0}^{0.5} \int_{y=0}^x f(x, y) \, dy \, dx + \int_{x=0.5}^1 \int_{y=0}^{1-x} f(x, y) \, dy \, dx \\ &= 2 \int_{x=0}^{0.5} e^{-x} \left[\int_{y=0}^x e^{-y} \, dy \right] \, dx + 2 \int_{x=0.5}^1 e^{-x} \left[\int_{y=0}^{1-x} e^{-y} \, dy \right] \, dx = 1 - 2e^{-1} \end{aligned}$$

Example: Discrete-time RVs X and Y

Given $\Pr(X = x_i, Y = y_i) = 1/6, i = 1, \dots, 6$

with sample space $S_{XY} = \{(-2, 1), (1, -4), (5, -3), (-1, 2), (-3, -6), (4, 7)\}$

To find $\Pr(X^2 + Y^2 < 20)$,

$$\Pr(X^2 + Y^2 < 20) = \Pr((x, y) \in D) = \sum_{(x_i, y_i) \in D} \Pr(X = x_i, Y = y_i)$$

Region D : points in intersection of S_{XY} and $x^2 + y^2 < 20$ (circle of radius $\sqrt{20} = 4.47$)

$$\begin{aligned} \Pr(X^2 + Y^2 < 20) &= \Pr(X = -2, Y = 1) + \Pr(X = 1, Y = -4) + \Pr(X = -1, Y = 2) \\ &= 3/6 \end{aligned}$$